Functional Requirements Document
For
Restaurant Management

Introduction

Project Overview:

The Restaurant Management System (RMS) is a cloud-based platform that integrates ordering, reservations, inventory, staff management, payments, and reporting into one system for web and mobile use. It helps restaurants improve efficiency, reduce costs, and enhance customer experience through real-time data and automation. With features like online ordering, table booking, POS and kitchen integration, and business analytics, RMS enables smarter decision-making and supports business growth through a secure and scalable SaaS model.

Project Goal:

1. Improve Order Management
The system will allow customers to place online orders and track them in real time while the kitchen and staff receive orders instantly.

2. Streamline Restaurant Operations
The system will manage table reservations, billing, kitchen orders, and staff work from one platform.

3. Enhance Customer Experience
Customers can book tables, place orders, earn loyalty points, and give feedback easily through the system.

4. Manage Inventory Efficiently
The system will track inventory levels, alert for low stock, and help in purchasing and costing recipes.

5. Increase Business Profitability
Sales, customer behavior, and inventory reports will help owners improve performance and profits.

6. Improve Staff Management
The system will manage staff schedules, roles, and communication for smoother operations.

7. Ensure Data Security and Reliability
The system will use secure login, data encryption, and backups to keep all information safe.

Target Audience:

- Restaurant owners and managers who want to automate daily work
- Small cafés, pizza shops, and fast-food outlets
- Medium and large restaurants with many tables and staff
- Cloud kitchens and online food delivery businesses

Key Features:

1. Order Management

- Online food ordering via web & mobile
- Table reservation system with time slots & waitlist
- Real-time order tracking
- Kitchen Display System (KDS)
- POS system integration

2. Inventory Management

- Real-time stock tracking
- Low-stock alerts
- Purchase order management
- Recipe costing & profitability analysis

3. Customer Relationship Management (CRM)

- Customer profiles & order history
- Loyalty programs (points, rewards, offers)
- Customer feedback & review management.

4. Reporting & Analytics

- Sales performance & trend reports
- Inventory usage & waste analysis
- Customer behavior & loyalty insights

5. Staff Management

- Employee scheduling & attendance tracking
- Payroll support

- Role-based access control
- Internal communication tools

6. User Interface & Experience

- Clean, intuitive UI
- Mobile-first design
- Personalized dashboards with KPIs

7. Integration & API

- Payment gateways integration
- Delivery service integration
- Open API for third-party/custom apps

8. Security & Data Privacy

- Data encryption
- Backup & disaster recovery
- GDPR / CCPA compliance

9. Scalability & Performance

- Cloud-based scalable architecture
- High performance & fast response times

10. Customer Support

- 24/7 multi-channel support
- Knowledge base & FAQs
- Regular updates & maintenance

11. SaaS Model

- Subscription-based pricing
- Feature-based plans
- Continuous upgrades

Problems:

1. Manual order and billing processes cause errors and delays.
2. Poor management of online orders and table reservations.
3. Lack of real-time order tracking between staff and kitchen.
4. Inefficient inventory control leading to wastage or stock shortages.
5. Difficulty tracking sales, profits, and performance.
6. Poor customer data, feedback, and loyalty management.
7. Manual staff scheduling and payroll issues.
8. Lack of data security, scalability, and system integration.

Opportunities:

1. **Online Ordering & Delivery:** Growing demand for food ordering through apps and websites.
2. **Cloud-based POS & Management Systems:** Restaurants need digital tools to manage sales, billing, and operations.
3. **Customer Loyalty & CRM:**Using data to attract, retain, and personalize service for customers.
4. **Inventory & Cost Control:** Software helps reduce food waste and increase profit.
5. **Analytics & Reporting:**Restaurants use data to improve pricing, menus, and staffing.
6. **Multi-channel Integration:** Linking delivery apps, payments, and marketing platforms in one system.
7. **Automation & Staff Management:**Scheduling, payroll, and kitchen display systems improve efficiency
8. **Subscription-based SaaS Solutions:** Selling restaurant management software as a monthly service is a fast-growing business.

System Overview

Purpose

The purpose of this Restaurant Management System is to streamline restaurant operations by integrating ordering, inventory, staff management, and customer engagement into one secure, scalable platform. It aims to improve efficiency, enhance customer experience, support data-driven decisions, and enable restaurants to grow through a reliable SaaS solution.

Scope

Key Functionalities:

1. **Order Management**
    - Online ordering via web and mobile
    - Table reservations with waitlist management
    - Real-time order tracking
    - Kitchen Display System (KDS) for staff
    - Integration with POS systems
2. **Inventory Management**
    - Real-time inventory tracking and alerts
    - Purchase order creation and management
    - Recipe costing and profitability analysis
3. **Customer Relationship Management (CRM)**
    - Customer profiles with order history and preferences
    - Loyalty programs with rewards and personalized offers
    - Feedback collection and analysis
4. **Reporting & Analytics**
    - Sales, inventory, and customer behavior reports
    - Insights to optimize operations and marketing
5. **Staff Management**
    - Employee scheduling, attendance tracking, and payroll
    - Role-based access control

- Internal communication tools

6. **User Interface & Experience**
    - Intuitive, mobile-first design
    - Personalized dashboards with key performance indicators (KPIs)
7. **Integration & API**
    - Payment gateways, delivery platforms, and marketing tools
    - API for custom application integration
8. **Security & Data Privacy**
    - Data encryption and secure storage
    - Backup, recovery, and compliance with regulations
9. **Scalability & Performance**
    - Scalable infrastructure for growth
    - Fast, reliable performance across devices
10. **Customer Support**
    - 24/7 support, knowledge base, and FAQs
    - Regular updates and system maintenance
11. **SaaS Model**
    - Subscription-based pricing
    - Continuous feature updates and enhancements

System Context

1.Order & Reservation Management – Supports online ordering, table reservations, real-time order tracking, kitchen display systems (KDS), and POS integration.

2.Inventory Management – Tracks stock levels in real-time, manages purchase orders, and calculates recipe costs to optimize profitability.

3.Customer Relationship Management (CRM) – Stores customer profiles, manages loyalty programs, collects feedback, and provides personalized offers.

4.Reporting & Analytics – Generates sales, inventory, and customer behavior reports to aid decision-making.

5.Staff Management – Handles employee scheduling, attendance tracking, role-based permissions, and internal communication.

6.User Interface & Experience – Mobile-first, intuitive UI with personalized dashboards for different user roles.

7.Integration & API – Supports third-party integrations (payment gateways, delivery services, marketing tools) and provides APIs for custom extensions.

8.Security & Compliance – Data encryption, backup & recovery, and adherence to regulations like GDPR/CCPA.

9.Scalability & Performance – Designed to scale with business growth while ensuring fast and smooth performance.

10.Customer Support & SaaS Model – Offers 24/7 support, knowledge base, regular updates, and subscription-based pricing with continuous feature upgrades.

System Architecture

1. Presentation Layer (UI/UX)

Handles all interactions with users—both customers and staff.

- **Web & Mobile Interfaces:** Clean, intuitive, mobile-first design.
- **Personalized Dashboards:** KPI display for managers, order views for staff, loyalty info for customers.
- **Communication Tools:** Chat or notifications between staff and customers.

2. Application/Service Layer (Business Logic)

Processes the core functionality and rules of the system.

- **Order Management:**
    - Online ordering, table reservations, order tracking, KDS integration, POS communication.
- **Inventory Management:**
    - Real-time tracking, purchase order management, recipe costing.
- **CRM (Customer Relationship Management):**
    - Customer profiles, loyalty programs, feedback analysis.
- **Staff Management:**
    - Employee scheduling, role-based permissions, communication tools.

- **Reporting & Analytics:**
    - Sales, inventory, and customer behavior reports.

3. Integration Layer

Handles communication with external systems and APIs.

- **Third-Party Integrations:** Payment gateways, delivery services, marketing platforms.
- **Internal APIs:** Expose services for internal or third-party developers.
- **POS Integration:** Sync orders, payments, and inventory with existing systems.

4. Data Layer (Persistence)

Manages storage, retrieval, and protection of data.

- **Database Management:** Customer data, orders, inventory, staff info.
- **Data Security & Privacy:** Encryption, compliance with GDPR/CCPA.
- **Backup & Recovery:** Regular data backups and disaster recovery planning.

5. Infrastructure Layer

Supports scalability, performance, and reliability.

- **Scalable Hosting:** Cloud-based servers to handle growth.
- **High Performance:** Fast loading, efficient queries, caching strategies.
- **Maintenance & Updates:** Regular software upgrades and patches.

Comprehensive Functional Requirements

4.1 User Interface Requirements

The system shall provide a user-friendly, intuitive, and responsive interface accessible through both web and mobile platforms. The following UI requirements must be met:

1. The system shall provide a login and authentication interface for different user roles, including customers, staff, and administrators.

2. The system shall display a personalized dashboard based on the user's role (e.g., customer dashboard, staff dashboard, manager dashboard).

3. The system shall allow customers to:
    - View menus with item descriptions, prices, and availability.
    - Place online orders and track order status in real time.
    - Make, modify, or cancel table reservations.

4. The system shall allow staff to:
    - View and manage incoming orders.
    - Update order status (e.g., received, preparing, ready, served).
    - Access inventory and kitchen display interfaces.
5. The system shall allow administrators to:
    - Manage menu items, prices, and availability.
    - Generate reports and analytics.
    - Configure system settings and user permissions.
6. The interface shall be responsive and support multiple devices and screen sizes.
7. The system shall provide clear notifications, alerts, and confirmation messages for user actions.

4.2 Data Flow Diagrams (DFDs)

The Data Flow Diagrams describe how data moves through the system and how different processes interacts

1. External Entities

External entities are people or systems outside the restaurant system that interact with it.

1. Customer

- Sends **order requests**
- Sends **reservation requests**
- Gives **customer information and feedback**

2. Administrator

- Sends **purchase orders**
- Receives **reports and analytics**

3. Staff

- Enters **sales details**
- Manages orders using the POS system

4. Payment Gateway

- Receives **payment details**
- Sends **payment confirmation**

5. POS Integration System

- Sends **POS orders**

- Shares **sales data**

2. Sub-Processes

Sub-processes are the main activities done by the system.

1. Order Management

- Receives orders from customers
- Processes and confirms orders
- Sends orders to the kitchen

2. Reservation Management

- Receives reservation requests
- Checks table availability
- Confirms reservations

3. CRM (Customer Management)

- Stores customer details
- Stores customer feedback

4. Kitchen Display Processing

- Shows confirmed orders to kitchen staff

5. Payment Processing

- Sends payment request to payment gateway
- Receives payment confirmation

6. Reporting and Analytics

- Creates sales and performance reports
- Sends reports to administrator

3. Data Stores

Data stores are places where data is saved.

1. Orders

- Stores order details like items, quantity, and price

2. Reservations

- Stores reservation details like date, time, and table number

3. CRM Database

- Stores customer information and feedback

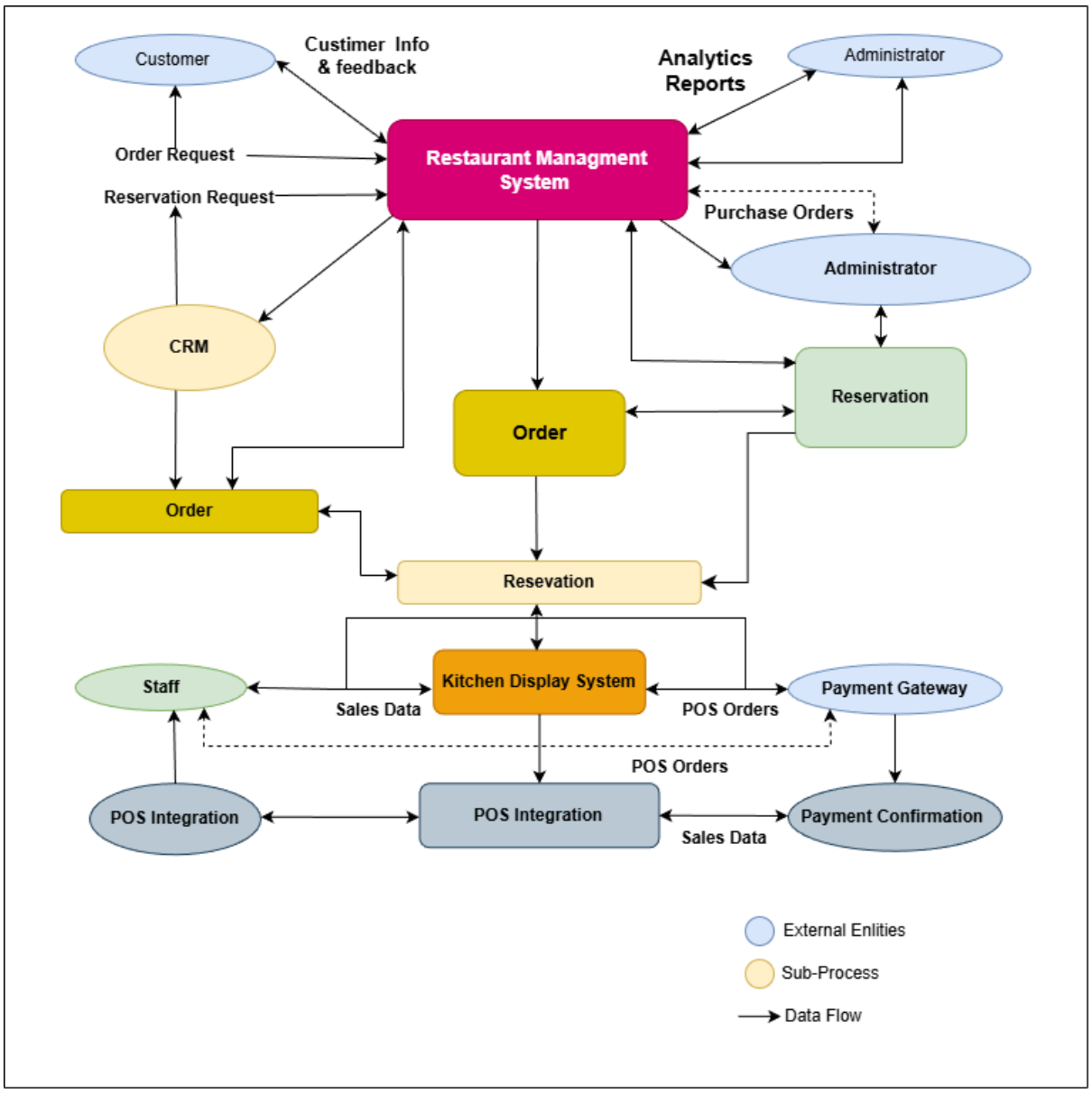
4. Sales Data

- Stores payment and transaction details

4. Data Flows

Data flows show how data moves in the system.

- **Order Request** flows from Customer to Order Management
- **Reservation Request** flows from Customer to Reservation Management
- **Customer Information & Feedback** flows from Customer to CRM
- **Order Details** flow from Order Management to Orders data store
- **Reservation Details** flow from Reservation Management to Reservations data store
- **Confirmed Orders** flow to Kitchen Display System
- **Payment Details** flow from system to Payment Gateway
- **Payment Confirmation** flows from Payment Gateway to system
- **Sales Data** flows from Staff and POS Integration to Sales Data store
- **Reports** flow from Reporting process to Administrator



5. Conclusion

The Data Flow Diagram of the Restaurant Management System clearly shows how orders, reservations, payments, and reports move through the system. It helps in smooth restaurant operations and better customer service.

4.3 Use Case Diagrams

The use case diagrams illustrate the interactions between system users (actors) and the system.

🎭 Actors

- Customer
- Restaurant
- Staff
- Kitchen
- Staff
- Manager / Admin
- Supplier
- Payment Gateway (external system)
- Delivery Service (external system.

Use Cases

👤 Customer

- Register / Login
- View Menu
- Place Online Order
- Make Table
- ReservationTrack
- OrderMake Payment
- Give Feedback
- View Loyalty Points

🍽️ Restaurant Staff

- Manage Orders
- Update Order
- StatusManage
- Reservations
- Handle POS

- Communicate with Kitchen

👨‍🍳 Kitchen Staff

- View Orders (KDS)
- Prepare Food
- Update Order Status

🧑‍💼 Manager / Admin

- Manage Menu
- Manage Inventory
- Manage Staff
- Generate Reports
- Manage Loyalty ProgramsManage Customer ProfilesConfigure System & Roles

🚚 Supplier

- Receive Purchase Orders
- Update Delivery Status

💳 External Systems

- Process Payment (Payment Gateway)Handle Delivery (Delivery Service)

4.4 Detailed Functional Specifications

4.4.1 Order Management

- The system shall allow customers to place orders online via the website or mobile application.
- The system shall route confirmed orders to the kitchen display system.
- The system shall allow staff to update order status in real time.
- The system shall integrate with POS systems for billing and order synchronization.

4.4.2 Reservation Management

- The system shall allow customers to reserve tables for specific dates and time slots.
- The system shall maintain a waitlist when tables are unavailable.
- The system shall notify customers of reservation confirmations or changes.

4.4.3 Inventory Management

- The system shall track inventory levels in real time.
- The system shall generate low-stock alerts automatically.
- The system shall allow administrators to create and manage purchase orders.
- The system shall support recipe costing and inventory usage tracking.

4.4.4 Customer Relationship Management (CRM)

- The system shall store customer profiles, order history, and preferences.
- The system shall support loyalty programs, including points and rewards.
- The system shall collect customer feedback and reviews.

4.4.5 Reporting and Analytics

- The system shall generate sales, inventory, and customer behavior reports.
- The system shall allow administrators to view performance metrics and trends.
- The system shall support data export for business analysis.

4.4.6 Staff Management

- The system shall allow administrators to create staff schedules and track attendance.
- The system shall support role-based access control.
- The system shall facilitate internal communication between management and staff

Non-Functional Requirements

1. Performance Requirements

- **Response Time:** System should load pages within 2 seconds.
- **Scalability:** Must handle increasing users, orders, and data efficiently.
- **Throughput:** Should process a high number of transactions per minute without delay.

2. Security Requirements

- **Authentication:** Secure login for staff and customers.
- **Authorization:** Role-based access control for admin, staff, and customers.
- **Data Encryption:** Sensitive data must be encrypted during storage and transmission.

3. Availability & Reliability

- **Availability:** System should be available 24/7 with minimal downtime.
- **Fault Tolerance:** System should recover automatically from failures.
- **Backup & Recovery:** Regular data backups and disaster recovery mechanisms must be implemented.

4. Usability Requirements

- **User-Friendly Interface:** Easy to navigate for customers and staff.
- **Mobile Compatibility:** Must support mobile, tablet, and desktop devices.
- **Accessibility:** System should be easy to learn with minimal training.

5. Maintainability

- **Modular Design:** System should be easy to modify and extend.
- **Documentation:** Proper technical and user documentation must be provided.
- **Bug Fixing:** Easy to update and fix system errors.

6. Compatibility & Portability

- **Cross-Platform Support:** Works on different browsers and operating systems.
- **Integration:** Must integrate with POS systems, payment gateways, and delivery platforms.

7. Compliance Requirements

- **Data Privacy:** Must comply with data protection laws (e.g., GDPR, local regulations).
- **Audit Logs:** Maintain logs for security and compliance tracking.

Assumptions

1. Business Assumptions

- Restaurants have stable internet connectivity.
- Users are familiar with basic digital systems.
- Restaurants agree to use a SaaS subscription model.

2. Functional Assumptions

- All restaurants follow standard order and billing processes.
- Inventory data is updated regularly by staff.
- POS systems used by restaurants support integration.

3. User Assumptions

- Customers can access the system via web or mobile devices.
- Staff roles (admin, cashier, kitchen staff) are clearly defined.
- Users will provide accurate information during registration.

4. Technical Assumptions

- System is hosted on a scalable cloud infrastructure.
- Third-party APIs (payment, delivery) are reliable and available.
- Browsers and devices used are up to date.

5. Security Assumptions

- Users keep their login credentials confidential.
- Restaurant management follows basic security best practices.

6. Operational Assumptions

- Regular system maintenance will be scheduled.
- Data backups are performed automatically.
- Customer support is available 24/7.

7. Regulatory Assumptions

- Restaurants comply with local food and business regulations.
- Data privacy laws applicable to the region are followed.

Acceptance Criteria

1. Order Management

- Users can place, edit, and cancel orders (web/mobile).
- Table reservations can be created with time slots and waitlist.
- Order status updates in real time (Placed → Preparing → Served/Delivered.
- The Kitchen receives orders instantly via KDS.
- POS syncs orders and payments without errors.

2. Inventory Management

- The system shows real-time stock levels.
- Low-stock alerts are triggered automatically.
- Users can create and track purchase orders.
- Recipe costing calculates item-wise cost correctly.
- Inventory reports can be generated and exported.

3. CRM (Customer Management)

- Customer profiles store details and order history.
- Loyalty points are added/redeemed correctly.
- The system can send personalized offers.
- Feedback and ratings can be submitted and viewed.

4. Reporting & Analytics

- Sales reports show daily/weekly/monthly data.
- Inventory usage and wastage reports are available.
- Customer behavior reports can be generated.
- Reports load within acceptable time and are accurate.

5. Staff Management

- Admin can create and manage staff schedules.
- Role-based access controls restrict features correctly.
- Attendance and basic payroll data can be tracked.
- Internal communication messages can be sent/received.

6. UI / UX

- Interface is responsive on web and mobile.
- Navigation is simple and consistent.
- Each user sees a role-based dashboard.
- Key actions are reachable within 2–3 clicks.

7. Integration & API

- Payment gateways process transactions successfully.
- Delivery and third-party services sync correctly.
- APIs return valid responses with proper authentication.
- Failed integrations show clear error messages.

8. Security & Data

- Sensitive data is encrypted in storage and transit.
- Regular backups can be restored successfully.
- Only authorized users can access protected data.
- The system complies with basic data privacy rules.

9. Performance & Scalability

- Pages load within acceptable time limits.
- The system supports multiple users concurrently.
- No data loss during high load.
- New outlets/users can be added without downtime.

10. Support & SaaS

- Users can raise support tickets successfully.
- FAQs and help content are accessible.
- Subscription plans can be created/updated.
- The System receives regular updates without breaking features.

Non-Functional Acceptance Criteria:

1. Performance

- The system shall load any main screen (Dashboard, Orders, Inventory) within ≤ 3 seconds under normal load.
- The system shall process and confirm an order within ≤ 2 seconds after submission.
- The system shall support at least 500 concurrent users per restaurant without performance degradation.
- Real-time order status updates shall be reflected within ≤ 1 second.

2. Availability & Reliability

- The system shall maintain 99.9% uptime per month, excluding planned maintenance.
- Scheduled maintenance shall not exceed 2 hours per month and must be communicated at least 24 hours in advance.
- In case of system failure, the system shall recover and resume operations within 15 minutes.

3. Scalability

- The system shall scale horizontally to support:
    - An increase of 2× users without code changes.
    - Multiple restaurant branches under a single account.
- Adding a new restaurant location shall not impact existing restaurant performance.

4. Security

- All sensitive data shall be encrypted in transit (TLS 1.2+) and at rest (AES-256).
- The system shall enforce role-based access control (RBAC) for all users.
- Users shall be automatically logged out after 15 minutes of inactivity.
- The system shall log all critical actions (logins, payments, refunds, inventory changes).
- The system shall pass OWASP Top 10 vulnerability checks.

5. Data Privacy & Compliance

- The system shall comply with GDPR and CCPA requirements.
- Users shall be able to request data export or deletion within 30 days.
- Customer personal data shall not be shared with third parties without explicit consent.

6. Usability & Accessibility

- A new user shall be able to complete a basic task (place an order, reserve a table) within ≤ 3 steps.
- The UI shall be responsive and usable on mobile, tablet, and desktop devices.
- The system shall conform to WCAG 2.1 Level AA accessibility standards.
- Error messages shall be clear, user-friendly, and actionable.

7. Compatibility

- The system shall support the latest 2 versions of major browsers (Chrome, Firefox, Safari, Edge).
- Mobile applications shall support Android and iOS current and previous major versions.
- POS integration shall work with supported third-party systems without data loss.

8. Maintainability

- The system shall allow deployment of bug fixes without downtime.
- Code changes shall not break existing features (verified by automated regression tests).
- System configuration changes shall not require code changes.

9. Backup & Recovery

- The system shall perform daily automated backups.
- Backup data shall be retained for at least 30 days.
- Data restoration shall be completed within 1 hour of a recovery request.

10. Support & Monitoring

- System health (CPU, memory, response time, errors) shall be monitored 24/7.
- Critical incidents shall trigger alerts within 5 minutes.
- Customer support requests shall receive an initial response within 1 hour.10

Additional Considerations:

- **Data Migration:** Support import of existing customer, menu, and inventory data from old systems.
- **Multi-Outlet Support:** Allow managing multiple restaurant branches from one account.
- **Localization:** Support multiple languages, currencies, taxes, and regional formats.
- **Offline Mode:** Basic POS and order taking should work during internet outages and sync later.
- **Audit Logs:** Maintain logs for critical actions (orders, refunds, inventory changes, user access).
- **Customization:** Allow restaurants to customize menus, workflows, tax rules, and reports.
- **Training & Onboarding:** Provide tutorials, demos, and onboarding guides for new users.
- **Accessibility:** Ensure usability for differently-abled users (keyboard support, readable UI, contrast).
- **Legal & Tax Compliance:** Support local tax rules, invoicing standards, and regulatory reporting.
- **Monitoring & Alerts:** System should notify admins about failures, low stock, or unusual activity.
- **Version Control & Rollback:**Ability to roll back in case of failed updates or releases.
- **Vendor Lock-in Avoidance:** Provide data export options in standard formats (CSV, Excel, etc.).

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Future Considerations

- **AI & Predictive Analytics**: Use AI to forecast demand, optimize inventory, and provide sales predictions.
- **Advanced Personalization:** Offer personalized menus, promotions, and recommendations based on customer behavior.
- **Multi-Outlet Support**: Enable centralized management for restaurant chains and franchises.
- **Third-Party Expansion**: Support additional integrations with new delivery partners and payment gateways.
- **Voice & Chat Ordering:** Enable voice assistants and chatbot-based ordering systems.
- **Enhanced Reporting**: Introduce real-time dashboards and advanced business intelligence tools.
- **Globalization Support:** Add multi-language and multi-currency capabilities for international expansion.
- **Automation Enhancements**: Automate supplier ordering, staff scheduling, and kitchen workflows.